

Modal question bank for 2nd unit

Problem 1

Let $S = \{(1, 2), (1, 3), (2, 1), (3, 3)\}$. Find S^2 (the composition $S \circ S$).

By definition $S^2 = \{(a, c) \mid \exists b \text{ with } (a, b) \in S \text{ and } (b, c) \in S\}$. We check all possible intermediate b :

- From $(1, 2) \in S$: take $b = 2$. Pairs in S with first coordinate 2 are $(2, 1)$, so $(1, 1) \in S^2$.
- From $(1, 3) \in S$: take $b = 3$. Pairs in S with first coordinate 3 are $(3, 3)$, so $(1, 3) \in S^2$.
- From $(2, 1) \in S$: take $b = 1$. Pairs in S with first coordinate 1 are $(1, 2), (1, 3)$, so $(2, 2), (2, 3) \in S^2$.
- From $(3, 3) \in S$: take $b = 3$. Pairs in S with first coordinate 3 are $(3, 3)$, so $(3, 3) \in S^2$.

Hence

$$S^2 = \{(1, 1), (1, 3), (2, 2), (2, 3), (3, 3)\}.$$

Problem 2

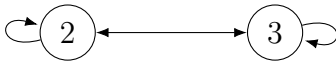
Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 1), (1, 4), (4, 1), (4, 4), (2, 2), (2, 3), (3, 2), (3, 3)\}$ on A . Draw its matrix and graph.

Matrix of R

Index rows and columns in order 1, 2, 3, 4. The adjacency matrix M_R has entry $m_{ij} = 1$ if $(i, j) \in R$, otherwise 0.

$$M_R = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 1 & 0 & 0 & 1 \\ 2 & 0 & 1 & 1 & 0 \\ 3 & 0 & 1 & 1 & 0 \\ 4 & 1 & 0 & 0 & 1 \end{array}$$

Graph of R (directed)



Problem 3

Define Monoid.

A *monoid* is an algebraic structure $(M, \cdot)$ consisting of a set M together with a binary operation $\cdot : M \times M \rightarrow M$ such that:

1. (Associativity) For all $a, b, c \in M$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
2. (Identity element) There exists an element $e \in M$ such that for all $a \in M$, $e \cdot a = a = a \cdot e$.

Example: $(\mathbb{N}_0, +)$ (natural numbers including 0, with addition) is a monoid with identity 0.

Problem 4

Define Group with respect to any binary operation.

A *group* is a set G together with a binary operation $*$: $G \times G \rightarrow G$ satisfying:

1. (Associativity) $\forall a, b, c \in G : (a * b) * c = a * (b * c)$.
2. (Identity) $\exists e \in G$ such that $\forall a \in G : e * a = a = a * e$.
3. (Inverse) $\forall a \in G$ there exists $a^{-1} \in G$ with $a * a^{-1} = e = a^{-1} * a$.

If the operation is commutative ($a * b = b * a$ for all a, b), the group is called *abelian*.

Problem 5

If R and S are equivalence relations on set A , prove that $R \cap S$ is an equivalence relation.

Proof. We show $R \cap S$ is reflexive, symmetric, transitive.

Reflexive. Since R and S are equivalence relations,

$\forall a \in A$ we have $(a, a) \in R$ and $(a, a) \in S$.

Hence $(a, a) \in R \cap S$. So $R \cap S$ is reflexive.

Symmetric. Suppose $(a, b) \in R \cap S$.

Then $(a, b) \in R$ and $(a, b) \in S$. Because R, S are symmetric, $(b, a) \in R$ and $(b, a) \in S$.

Therefore $(b, a) \in R \cap S$.

Transitive. Suppose $(a, b) \in R \cap S$ and $(b, c) \in R \cap S$.

Then $(a, b), (b, c) \in R$ and also $\in S$.

Since R is transitive, $(a, c) \in R$. Since S is transitive, $(a, c) \in S$.

Thus $(a, c) \in R \cap S$.

Having all three properties, $R \cap S$ is an equivalence relation. □

Problem 6

Verify whether the relation

$$R = \{(1, 1), (1, 4), (4, 1), (2, 2), (2, 3), (3, 4), (3, 3), (3, 2), (4, 3), (4, 4)\}$$

on $X = \{1, 2, 3, 4\}$ is an equivalence relation.

Check properties:

- **Reflexive:** $(1, 1), (2, 2), (3, 3), (4, 4)$ are all present, so reflexive holds.
- **Symmetric:** For each pair:
 - $(1, 4)$ and $(4, 1)$ both present.
 - $(2, 3)$ and $(3, 2)$ both present.
 - $(3, 4)$ and $(4, 3)$ both present.

So symmetry holds.

- **Transitive:** need for all $(a, b), (b, c) \in R$ that $(a, c) \in R$. Consider $(1, 4) \in R$ and $(4, 3) \in R$. If transitive, we should have $(1, 3) \in R$. But $(1, 3) \notin R$. Thus transitivity fails.

Therefore R is not an equivalence relation (counterexample: $(1, 4), (4, 3) \in R$ but $(1, 3) \notin R$).

Problem 7

Show that “greater than or equal” ($\geq$) is a partial ordering on the set of integers $\mathbb{Z}$.

Proof. We check reflexivity, antisymmetry, transitivity on $\mathbb{Z}$.

- Reflexive: For any $a \in \mathbb{Z}$, $a \geq a$ holds.
- Antisymmetric: If $a \geq b$ and $b \geq a$, then $a = b$ (true for integers).
- Transitive: If $a \geq b$ and $b \geq c$ then $a \geq c$.

Hence $(\mathbb{Z}, \geq)$ is a partial order. □

Problem 8

Let $X = \{1, 2, 3, 4, 5, 6, 7\}$ and define $R = \{(x, y) \in X \times X \mid x - y \text{ is divisible by } 3\}$. Show R is an equivalence relation.

Answer :

Reflexive: For any x , $x - x = 0$ divisible by 3, so $(x, x) \in R$.

Symmetric: If $x - y$ is divisible by 3, then $y - x = -(x - y)$ is also divisible by 3, so $(y, x) \in R$.

Transitive: If $x - y \equiv 0 \pmod{3}$ and $y - z \equiv 0 \pmod{3}$, then $x - z \equiv (x - y) + (y - z) \equiv 0 \pmod{3}$, so $(x, z) \in R$.

Thus R is an equivalence relation. The equivalence classes (modulo 3) intersected with X are:

$$[1] = \{1, 4, 7\}, \quad [2] = \{2, 5\}, \quad [3] = \{3, 6\}.$$

Problem 9

Question (verbatim):

Define partition of a set and covering of sets, let

$$A_1 = \{\{a, b\}, \{b, c\}\}, \quad A_2 = \{\{a\}, \{a, c\}\}, \quad A_3 = \{\{a\}, \{b, c\}\},$$

$$A_4 = \{\{a, b, c\}\}, \quad A_5 = \{\{a\}, \{b\}, \{c\}\}, \quad A_6 = \{\{a\}, \{a, b\}, \{a, c\}\}.$$

Which of the following sets are partitions and covers the set $A = \{a, b, c\}$.

Answer:

Definition (Covering of a Set). Let A be a non-empty set and let $\mathcal{C} = \{C_i \mid i \in I\}$ be a collection of non-empty subsets of A , where I is an index set. We say that $\mathcal{C}$ is a *covering* (or *cover*) of A if

$$\bigcup_{i \in I} C_i = A.$$

In other words, every element of A belongs to at least one subset in the collection $\mathcal{C}$. Note that in a covering, subsets may overlap (i.e., they need not be disjoint).

Definition (Partition of a Set). Let A be a non-empty set and let $\mathcal{P} = \{P_i \mid i \in I\}$ be a collection of non-empty subsets of A . We say that $\mathcal{P}$ is a *partition* of A if the following two conditions are satisfied:

1. **Covering condition:**

$$\bigcup_{i \in I} P_i = A.$$

That is, every element of A belongs to at least one subset P_i .

2. **Mutual disjointness:** For all $i, j \in I$ with $i \neq j$, we have

$$P_i \cap P_j = \emptyset.$$

That is, no element of A appears in more than one subset.

Check each collection:

1. $A_1 = \{\{a, b\}, \{b, c\}\}$.

- Union: $\{a, b\} \cup \{b, c\} = \{a, b, c\} = A$. Hence A_1 is a **cover**.
- Intersection: $\{a, b\} \cap \{b, c\} = \{b\} \neq \emptyset$. So blocks are not pairwise disjoint; A_1 is **not a partition**.

2. $A_2 = \{\{a\}, \{a, c\}\}$.

- Union: $\{a\} \cup \{a, c\} = \{a, c\}$ which does **not** include b . So A_2 is **not a cover** and thus **not a partition**.

3. $A_3 = \{\{a\}, \{b, c\}\}$.

- Union: $\{a\} \cup \{b, c\} = \{a, b, c\} = A$. So A_3 is a **cover**.
- Intersection: $\{a\} \cap \{b, c\} = \emptyset$ and both blocks are nonempty. Therefore A_3 is a **partition**.

4. $A_4 = \{\{a, b, c\}\}$.

- Union: $\{a, b, c\} = A$ so A_4 is a **cover**.

- There is only one (nonempty) block, so pairwise disjointness holds vacuously; A_4 is a **partition** (the trivial partition).

5. $A_5 = \{\{a\}, \{b\}, \{c\}\}$.

- Union: $\{a\} \cup \{b\} \cup \{c\} = A$, so A_5 is a **cover**.
- The three sets are pairwise disjoint and nonempty; A_5 is a **partition**.

6. $A_6 = \{\{a\}, \{a, b\}, \{a, c\}\}$.

- Union: $\{a\} \cup \{a, b\} \cup \{a, c\} = \{a, b, c\} = A$, so A_6 is a **cover**.
- Blocks overlap (for example $a \in \{a\} \cap \{a, b\}$), therefore A_6 is **not a partition**.

Summary table:

Collection	Is a cover?	Is a partition?
A_1	Yes	No
A_2	No	No
A_3	Yes	Yes
A_4	Yes	Yes
A_5	Yes	Yes
A_6	Yes	No

10. Determine whether the binary operation $*$ is commutative and associative on the set

$$(i) \mathbb{Z}_+, \quad a * b = a + b + 2. \quad (ii) \mathbb{Z}, \quad a * b = ab.$$

Answer (i)

Commutativity: For all $a, b \in \mathbb{Z}_+$,

$$a * b = a + b + 2 = b + a + 2 = b * a,$$

so $*$ is *commutative*.

Associativity: Compute

$$(a * b) * c = (a + b + 2) + c + 2 = a + b + c + 4,$$

and

$$a * (b * c) = a + (b + c + 2) + 2 = a + b + c + 4.$$

Hence $(a * b) * c = a * (b * c)$ for all a, b, c , so $*$ is *associative*.

Remark: Although associative and commutative, $\mathbb{Z}_+$ with this operation does not have an identity in $\mathbb{Z}_+$ because solving $a * e = a$ gives $e = -2 \notin \mathbb{Z}_+$.

Answer (ii)

On $\mathbb{Z}$ define $a * b = ab$ (ordinary integer multiplication).

Commutativity: For all $a, b \in \mathbb{Z}$, $ab = ba$, so $*$ is *commutative*.

Associativity: Integer multiplication is associative: $(ab)c = a(bc)$, so $*$ is *associative*.

11. Let $A = \{2, 3, 6, 12, 24, 36\}$ and the relation on A defined as $a \leq b$ if a divides b . Draw the Hasse diagram for $(A, \leq)$.

Answer

We first determine the cover relations (i.e. $x \prec y$ when $x \mid y$ and there is no $z \in A$ with $x \mid z \mid y$ and $z \neq x, y$).

Divisibility relations among elements:

$$2 \mid 6, 2 \mid 12, 2 \mid 24, 2 \mid 36; \quad 3 \mid 6, 3 \mid 12, 3 \mid 24, 3 \mid 36;$$

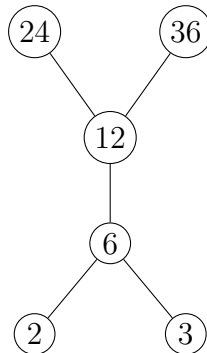
$$6 \mid 12, 6 \mid 24, 6 \mid 36; \quad 12 \mid 24, 12 \mid 36.$$

Covers (no intermediate in A):

$$2 \prec 6, \quad 3 \prec 6, \quad 6 \prec 12, \quad 12 \prec 24, \quad 12 \prec 36.$$

Note that 24 and 36 are distinct maximal elements and $24 \nmid 36$, $36 \nmid 24$.

Hasse diagram (levels bottom to top): bottom $\{2, 3\}$, then $\{6\}$, then $\{12\}$, top $\{24, 36\}$.



12. Show that $(\mathbb{Z}_5 \setminus \{0\}, \times_5)$ is an abelian group.

Answer

Let $G = \{1, 2, 3, 4\}$ be the nonzero residues modulo 5, with operation multiplication modulo 5 (denote $\times_5$).

- **Closure:** If $a, b \in G$, then $ab \not\equiv 0 \pmod{5}$ (since 5 is prime), so $ab \pmod{5} \in G$.
- **Associativity:** Follows from associativity of integer multiplication; reduction mod 5 respects the product.
- **Identity:** 1 is the identity: $1 \cdot a \equiv a \pmod{5}$.
- **Inverses:** Each element has a multiplicative inverse mod 5:

$$1^{-1} = 1, \quad 2^{-1} = 3 \text{ (since } 2 \cdot 3 = 6 \equiv 1), \quad 3^{-1} = 2, \quad 4^{-1} = 4 \text{ (} 4 \cdot 4 = 16 \equiv 1).$$

- **Commutativity:** Multiplication mod 5 is commutative.

Thus $(\mathbb{Z}_5^\times, \times_5)$ is an abelian group.

13. Show that $(\mathbb{Z}, *)$ is an abelian group with respect to the binary operation $a * b = a + b + 1$.

Answer

Let $*$ be defined on $\mathbb{Z}$ by $a * b = a + b + 1$.

- **Closure:** $a + b + 1 \in \mathbb{Z}$ for all $a, b \in \mathbb{Z}$.
- **Associativity:** For all $a, b, c \in \mathbb{Z}$,

$$(a * b) * c = (a + b + 1) + c + 1 = a + b + c + 2 = a + (b + c + 1) + 1 = a * (b * c).$$

- **Identity:** Find $e \in \mathbb{Z}$ with $a * e = a$ for all a . Solve $a + e + 1 = a$, so $e = -1$. Check: $a * (-1) = a + (-1) + 1 = a$.
- **Inverses:** For given a , solve $a * b = e = -1$. That gives $a + b + 1 = -1 \Rightarrow b = -a - 2 \in \mathbb{Z}$. Thus every a has inverse $-a - 2$.
- **Commutativity:** $a * b = a + b + 1 = b + a + 1 = b * a$.

Therefore $(\mathbb{Z}, *)$ is an abelian group (identity -1 , inverse of a is $-a - 2$).

14. Let $S = \{1, 2, \dots, 10\}$ and relation R on S where $R = \{(x, y) \mid x + y = 10\}$. What are the properties of the relation?

Answer

List the pairs in R explicitly:

$$R = \{(1, 9), (2, 8), (3, 7), (4, 6), (5, 5), (6, 4), (7, 3), (8, 2), (9, 1)\}.$$

Check relation properties on S :

- **Reflexive:** A relation is reflexive if $(x, x) \in R$ for all $x \in S$. Here only $(5, 5) \in R$. Hence R is *not reflexive*.
- **Symmetric:** If $(x, y) \in R$ then $x + y = 10$ implies $y + x = 10$, so $(y, x) \in R$. Thus R is *symmetric*.
- **Transitive:** If $(x, y), (y, z) \in R$, then $x + y = 10$ and $y + z = 10$. Subtracting gives $x = z$. For (x, z) to be in R we need $x + z = 10$, hence $2x = 10$ which forces $x = 5$. So transitivity does not hold generally (only in trivial cases). Thus R is *not transitive*.
- **Antisymmetric:** There exist distinct x, y (e.g. $1, 9$) with $(1, 9), (9, 1) \in R$, so R is *not antisymmetric*.

Conclusion: R is *symmetric only* (neither reflexive, nor transitive, nor antisymmetric). It is not an equivalence relation.

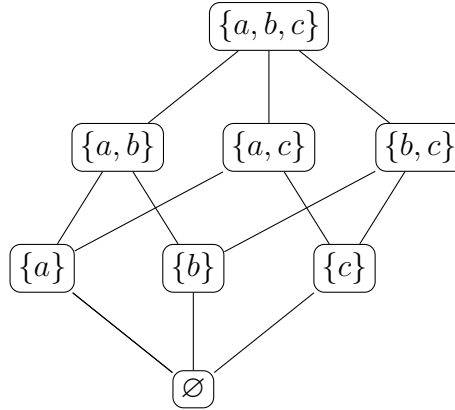
15. Draw the Hasse Diagram for the poset $(\mathcal{P}(S), \subseteq)$ where $S = \{a, b, c\}$.

Answer

The power set $\mathcal{P}(S)$ has the eight elements

$$\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}.$$

The poset under inclusion is the Boolean lattice of rank 3. The Hasse diagram (cover relations are between sets that differ by exactly one element) is shown below.



16. Let R and S be two relations on the set of positive integers

$$R = \{(x, 2x) \mid x \in \mathbb{Z}^+\}, \quad S = \{(x, 7x) \mid x \in \mathbb{Z}^+\}.$$

Find $R \circ R$, $R \circ S$, $R \circ R \circ R$, and $R \circ S \circ R$.

Answer

Interpret each relation as a function-like map:

$$R : x \mapsto 2x, \quad S : x \mapsto 7x.$$

Composition of relations corresponds to composition of these linear maps.

$$R \circ R = \{(x, 2(2x)) \mid x \in \mathbb{Z}^+\} = \{(x, 4x)\}.$$

$$R \circ S = \{(x, 2(7x)) \mid x \in \mathbb{Z}^+\} = \{(x, 14x)\}.$$

$$R \circ R \circ R = \{(x, 2(2(2x))) \mid x \in \mathbb{Z}^+\} = \{(x, 8x)\}.$$

$$R \circ S \circ R = \{(x, 2(7(2x))) \mid x \in \mathbb{Z}^+\} = \{(x, 28x)\}.$$

More generally, $R^n = \{(x, 2^n x)\}$ and compositions mixing R and S multiply by the corresponding constants in the order of composition.